

Bhoomi Sharma

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Motivated and eager AI/ML professional with foundational knowledge in data analysis, machine learning, and artificial intelligence. Skilled in applying statistical methods, data preprocessing, and basic machine learning algorithms to solve problems. Proficient in Python, with hands-on experience using libraries such as scikit-learn, Pandas, and NumPy. Enthusiastic about learning new AI techniques and contributing to innovative projects that leverage data-driven solutions.

SKILLS

- **Programming Languages and Libraries:** Python (pandas, scikit-learn, numpy, matplotlib, transformers, OpenAI API, Lang Chain, LangGraph)
- **Algorithms:** Linear Regression, Logistic Regression, Ridge/Lasso Regression, SVM, Decision Trees, Bagging and Boosting Techniques, Random Forest, k-Nearest Neighbors, K-Means, Hierarchical Clustering, BERT, Transformers
- **Data Science & Technologies:** Data science pipeline (cleaning, wrangling, visualization, modeling, interpretation, deployment), statistics, object-oriented programming (OOP), Git (GitHub, GitLab)
- **Data Visualization:** Excel, Matplotlib, Seaborn, Plotly
- **Databases & Search:** SQL (MySQL), vector databases (FAISS, Pinecone, PostgreSQL)
- **AI & Deep Learning:** Generative AI, CNN, YOLOv8 (Object Detection), PyTorch
- **Frameworks & Tools:** Django, OpenAI GPT-4, LangChain, LangGraph, Agentic AI

PROFESSIONAL EXPERIENCE

AI Developer

June 2025 – Present

(Alea Chatbot)

- Transitioned from intern to full-time developer, focusing on optimizing neural network architectures and deploying ML solutions.
- Implemented customized neural network architectures to optimize model performance and efficiency.
- Developed models for multi-regression problems.
- Applied machine learning and deep learning models based on research papers, evaluating their results to ensure optimal performance.
- Conducted graph data analysis and evaluated neural network models using various graph embedding techniques to gain enhanced insights and improve performance.

Projects at Alea:

- **Chatbot Development & Implementation:** Developed an advanced LLM-based chatbot capable of extracting and delivering precise information from multiple data sources, including PDFs, web scraping, and internal databases. Implemented NLP techniques to understand user queries and generate contextually relevant responses. Integrated data retrieval systems that search internal PostgreSQL databases and utilize web scraping methods. Enhanced responses with Google Search API and employed large language models like Google GenAI with LangChain and LangGraph. Utilized

Retrieval-Augmented Generation (RAG) techniques to improve the accuracy and relevance of generated responses by combining retrieval of relevant documents with generative models.

- **Threat and Face Detection:** Built object detection systems using YOLOv8 CNN trained on labelled datasets (e.g., weapons detection). Utilized PyTorch for model training and deployment. Developed real-time threat detection and face recognition solutions, integrated with backend APIs using Django. Leveraged OpenAI GPT-4 and Agentic AI for automated threat assessment and decision-making.
- **Jira Ticket Data Analysis & Prediction:** Analyzed data from Jira tickets to predict whether a new ticket was a bug or a task. Performed extensive data processing, feature engineering, and used DistilBERT for text classification. Developed a model that provides users with suggested solutions or support email contacts for bugs, enhancing support efficiency and user experience.

PROJECTS

- **Flight Delay Analysis**
Collected and integrated large-scale historical flight records and weather data (rain, wind, fog, thunderstorms), aligning them accurately by scheduled departure/arrival times and airport locations. Cleaned and preprocessed data by addressing missing or inconsistent entries, and standardizing date and time formats for accurate time-series analysis.
Conducted exploratory data analysis (EDA) to identify trends and peak delay periods, and used correlation analysis to evaluate the impact of weather conditions on delays.
Built and trained predictive models using regression and decision tree algorithms to forecast potential flight delays based on real-time weather forecasts.
- **Sales Performance Analysis**
Extracted, analyzed, and cleaned large-scale sales data to ensure accuracy and readiness for insights. Developed visualizations for key performance metrics, including revenue, sales volume, conversion rates, and customer acquisition costs.
Identified sales trends and seasonal patterns, enabling targeted regional and product-specific strategies to maximize revenue.
Presented comprehensive reports.

EDUCATION

- **Bachelor of Computer Applications (BCA)**
2023 - Present